CONSTRUCTIBLE SUBSETS PULL BACK ALONG A QUASI-COMPACT OPEN IMMERSION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The composition permanence of quasi-compactness. Invoked inside the counted proof (source lines 2854-2855). Makes retrocompact opens of the subspace retrocompact in the ambient space.

Source reference. Original lemma lemma-composition-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 1870–1870. Navigation pointer only: the counted text refers to the Stacks item lemma-composition-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $U \subset X$ be a retrocompact open. Let $E \subset U$. If E is constructible in U, then E is constructible in X.*

Proof. Suppose that $V, W \subset U$ are retrocompact open in U. Then V, W are retrocompact open in X (Lemma lemma-composition-quasi-compact). Hence $V \cap (U \setminus W) = V \cap (X \setminus W)$ is constructible in X. We conclude since every constructible subset of U is a finite union of subsets of the form $V \cap (U \setminus W)$. □

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